

Summit Fan

Boulder, Colorado
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Education

Bachelor of Science, Software Engineering, Mathematics, and Computer Science

May 2025

Rose-Hulman Institute of Technology, Terre Haute, IN

GPA: 3.98/4.00

Related Courses: Software Design, Software Quality Assurance, Debugging, Advanced Database Systems, Software Requirements Engineering, Object-Oriented Software Development, Data Structures and Algorithm Analysis, Software Construction and Evolution, Software Architecture, Software Program Understanding, Design and Analysis of Algorithms

Skills

Programming: Go, Java, Python, C, Racket/Scheme, SQL, JavaScript, Perl, C++, HTML, CSS

Tools: Git, Linux, GitLab, GitHub, Jira, Confluence, LaTeX, Slack, WSL, JUnit, Docker, Figma, Agile, Scrum, Redis, MongoDB, Neo4j, InfluxDB, Kafka, Markdown, GitHub Actions, AWS (SNS, SQS), Jenkins, Gradle, Windows, Kubernetes, Grafana, Google Workspace, Microsoft Office, Atlassian Suite, Microsoft SQL Server, Artifactory, Terraform

Personal: Learning, improving, teamwork, collaboration, problem-solving, curiosity

Experience

Current Work

June 2025 - present

- Continuing to learn Go and taking online computer science courses through CU Boulder
- Anticipating problems and next steps and improving communication skills through work as a stage technician
- Developing a walk scheduling mathematical model to improve my family's process for walking the dog
- Creating a portfolio website: <https://summit2021237.github.io>
- Biking up climbs around Boulder, Colorado

Cloud Software Engineering Intern, DEKA Research & Development Corp.

May 2024 - August 2024

- Developed software with CI/CD pipelines in a distributed microservice architecture using Kubernetes pods and RESTful APIs
- Drove a Go library from Jira ticket to release to forward and collect metrics, which has been used in multiple services
- Understood existing code to keep original functionality and remove duplication when designing the metrics library
- Started creating a Go service that polls events in a publish/subscribe system and then indexes them
- Wrote technical documentation, participated in code reviews, and contributed to daily stand-ups

Volunteer Work

- Raised over \$12,000 for the Courage Classic bike ride benefiting Children's Hospital Colorado going door-to-door
- Volunteered at bike races and repaired mountain biking trails
- Volunteered over 168 hours at FIRST robotics competitions and mentored three FIRST Tech Challenge teams

Leadership

- Principal bass in Flatirons Community Orchestra, All State Orchestra, youth orchestra, and the Rose-Hulman symphony

Project Experience

ROCKO, William A. Kline Innovation Award

September 2024 - May 2025

- Worked with a team to create a self-balancing wheeled robot for use as a mid-level robotics educational platform
- Designed and documented a ROS2-based, event-driven architecture in Python and C++ for modifiability and learnability
- Learned control theory and mathematics behind balancing bipedal robots

Get Attacked by a Bear

September 2024 - November 2024

- Developed a scalable, reliable system in Python with multiple NoSQL databases to recommend locations for bear attacks
- Incorporated MongoDB, Neo4j, and InfluxDB to track bear encounters, locations, population density, and reviews
- Used Kafka as a queue to process events and ensure databases do not lose data if they go down

Expanding and Documenting Catan

March 2025 - May 2025

- Improved a Java Catan game project by adding five new features and test cases for the new features
- Refactored code and test cases to be more understandable, testable, and maintainable
- Wrote user and developer guides, documenting installation, playing the game, CI/CD process, and design

Fixing Open-Source Software Defects

April 2024 - May 2024

- Collaborated with group members to fix two known defects with SageMath using a Python debugger
- Documented the process of finding the root cause of each defect and fixing the defect
- Verified that the defects were fixed

Honors and Activities

Flatirons Community Orchestra, <i>Principal Bass</i>	2025 - present	Road Cycling	2010 - present
Rose Symphony Orchestra, <i>Principal Bass and Orchestra Manager</i>	2021 - 2025	Jefferson Symphony Orchestra	2025 - present
Grand Prix Engineering Team, <i>Suspension Lead (2022-2024)</i>	2021 - 2024	Stage Technician	2022 - present
FIRST Robotics Competition, <i>Build Captain (2021)</i>	2017 - 2021	Combat Robotics Team	2021 - 2025
FIRST Tech Challenge, <i>Co-Captain (2017-2020)</i>	2016 - 2020	Rose Drama Club	2024 - 2025
Rose-Hulman Dean's List	All terms	Tau Beta Pi	2025 - present
International Baccalaureate Diploma	2017 - 2021	MATE Robotics Team	2021 - 2023

Hobbies: Road and gravel cycling, hiking and climbing, playing bass in orchestras, stage work, robotics, helping others improve